

25 Practice Questions

Step 1: Create the **t_employee** Table

First, let's create the **t_employee** table with some basic columns.

```
CREATE TABLE t_employee (  
    c_eid SERIAL PRIMARY KEY,  
    c_name VARCHAR(50),  
    c_department_id INT,  
    c_salary DECIMAL(10, 2),  
    c_hire_date DATE  
);
```

Step 2: Insert Sample Data into **t_employee** Table

```
INSERT INTO t_employee (c_name, c_department_id, c_salary, c_hire_date) VALUES  
( 'Alice', 1, 50000, '2020-01-15'),  
( 'Bob', 2, 60000, '2019-03-10'),  
( 'Carol', 3, 45000, '2021-07-23'),  
( 'David', 2, 70000, '2018-05-22'),  
( 'Eve', 4, 55000, '2017-08-19'),  
( 'Frank', 5, 48000, '2019-11-30'),  
( 'Grace', 5, 50000, '2021-01-01'),  
( 'Hannah', 5, 52000, '2020-02-17');
```

Step 3: Create the **t_department** Table

```
CREATE TABLE t_department (  
    c_department_id SERIAL PRIMARY KEY,  
    c_department_name VARCHAR(50) NOT NULL  
);
```

Step 4: Insert Sample Data into **t_department**

```
INSERT INTO t_department (c_department_name) VALUES  
( 'HR'),  
( 'Finance'),  
( 'Engineering'),  
( 'Marketing'),  
( 'Sales');
```

Attempt All Questions are compulsory

1. Use of **LIKE**

Q1. Find all employees whose name starts with 'A'.

Q2. Find employees whose names contain 'a'.

2. Use of **GROUP BY**

Q3. Get average salary per department.

3. Use of **HAVING**

Q4. Get departments with more than 1 employee.

4. Use of **BETWEEN**

Q5. Find employees hired between '2019-01-01' and '2020-12-31'.

5. Use of **IN**

Q6. List employees in departments 2, 4, or 5.

6. Use of **DISTINCT**

Q7. Show distinct salaries from employees.

7. Use of **UNIQUE** (on constraint)

Q8. Add a constraint to make employee names unique.

8. Use of **LIMIT**

Q9. Get top 3 highest paid employees.

9. Use of **UPDATE**

Q10. Increase salary of 'Alice' by 10%.

10. Use of **INSERT**

Q11. Add a new employee named 'Ivy'.

11. Use of **ALTER**

Q12. Add a new column for job title.

12. Use of **ALIAS**

Q13. Display employee name as "Employee Name".

Q14. Get department name and employee count using join and alias.

13. Combine Concepts

Q15. Get names of employees with salary between 48000 and 60000.

Q16. Get departments with total salary > 100000.

Q17. List distinct departments employees belong to.

Q18. Find all employees with names ending in 'e'.

Q19. Show employees with job title NULL.

Q20. Rename column **c_name** to **full_name**.

Q21. Find average salary for departments with more than 1 employee.

Q22. Get top 5 recently hired employees.

Q23. Insert employee with job title.

Q24. List all employees with salary not in (45000, 50000).

Q25. Find employees whose name is either 'Alice', 'Eve', or 'Grace'.